

Wellness Platform: Hybrid Dosha Classification and Weighted Centre Matching for Ayurveda and Spiritual Tourism in Sri Lanka

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Wellness Platform -- Tour Ceylon -- Unified Tourism Platform for Sri Lanka

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Abstract

This paper presents the wellness component of Tour Ceylon: a hybrid matcher that combines a Random Forest dosha (Vikruti) classifier with a transparent linear ranker over 83 curated centres (37 Curative, 32 Wellness/Rejuvenative, 14 Spiritual/Meditation). Tourists follow one of two paths--Ayurveda (69 centres, 12 conditions) or spiritual retreat (14 centres, 6 retreat types)--and complete a six-feature quiz (age, gender, sleep, appetite, temperature sensitivity, emotional tendency). The production classifier is `sklearn.ensemble.RandomForestClassifier` with `n_estimators = 100` and `random_state = 42`, trained at Flask import on 102 labelled profiles after dropping 8 Balanced (N/A) rows from a 110-row workbook (6 real Ayya's Case records and 104 expert-augmented rows). An offline six-model comparison file reports Random Forest accuracy 0.81, precision 0.832, recall 0.81, and F1-score 0.794, outperforming Decision Tree (F1 0.705), ANN (0.667), Naive Bayes (0.567), SVM (0.360), and k-NN (0.332). Centre score uses weights 0.35 condition, 0.20 dosha, 0.25 NLP quality/5, 0.20 budget. Ablation of those terms shows the full model average quality 4.484 versus 3.845 for condition-only. Timed on the development PC, `predict_dosha()` mean inference is 7.94 ms (n=200, p50 7.76 ms, p95 9.06 ms); joblib dump of the fitted RF is 406.97 KB. Outcome percentages in the UI are illustrative, not clinical claims. Limitations include the small real labelled set, 50/83 centres with unspecified dosha focus, and the absence of the original training notebook in the repository.

Index Terms-- Random Forest, Ayurveda, dosha classification, recommender systems, ablation study, scikit-learn, wellness tourism, Sri Lanka

I. INTRODUCTION

Sri Lanka offers Ayurvedic hospitals, wellness resorts, and meditation monasteries without a single tourist-facing matcher that jointly considers presenting condition, constitution (dosha), budget, and review-derived quality. The wellness component (Vite MPA, port 5181) fills that role inside the Tour Ceylon monolith. Machine learning is used only for dosha prediction. Centre ranking is a documented weighted sum so that an examiner can recompute a score by hand. Duration and outcome-likelihood maps are expert- and literature-informed lookups and are labelled in the UI as non-clinical.

II. RELATED WORK AND DATA SOURCES

Dosha (Vata, Pitta, Kapha) classification in software often uses questionnaires plus classical decision rules. This project replaces purely hand-coded rules with a Random Forest trained on a mixed real/augmented table, then keeps the ranker linear for auditability. Review quality is precomputed (`nlp_quality` on a 5-point scale) rather than inferred at request time by a transformer. Spiritual matching still runs dosha prediction even though many retreats are constitution-agnostic--an acknowledged limitation.

TABLE I
Centre pool (full_candidate_pool.json) and training workbook.

Item	Value
Centres (JSON bytes)	83 (102,918 bytes)
Curative / Wellness / Spiritual	37 / 32 / 14
Ayurveda path pool	69 (non-spiritual)
Dosha focus General/Unspecified	50 of 83
Dosha focus Vata / Pitta / Kapha / Balanced	11 / 3 / 3 / 16
Training workbook rows	110
Labelled after dropping Balanced (N/A)	102
Data origin REAL / AUGMENTED	6 / 104
Label counts before Balanced drop	Vata 53, Pitta 33, Kapha 16, Balanced 8
Features	6 (Age + 5 categorical)
sklearn version	1.5.1

III. METHODOLOGY

A. Technical Setup

Service: backend/app/services/matching_engine.py. Routes: /api/recommend, /api/match, admin CRUD and analytics. Persistence is dual: JSON is authoritative for matching; PostgreSQL tables wellness_centers, wellness_reviews, wellness_condition_mappings, and wellness_matching_sessions follow MVC. Admin password defaults to a shared demo secret overridable by WELLNESS_ADMIN_PASSWORD. Leaflet 1.9.4 renders maps. No GPU is used.

B. Pipeline

- Path selection: Ayurveda (12 conditions) or Spiritual (6 retreat types).
- Quiz POST /api/recommend: condition, age, gender, sleep_pattern, appetite, cold_heat_sensitivity, emotional_tendency, budget_tier, optional companion_condition, top_n=6.
- Dosha: LabelEncoder on five categoricals; Age numeric; RF predict + predict_proba; confidence = 100 * max(proba), one decimal.
- Unknown quiz categories map silently to the first encoder class.
- Path filter on category == Spiritual/Meditation versus the complement.
- Weighted score (Section III-D); companion blend 0.65 primary + 0.35 companion.
- Keyword risk scan on Google-style review strings.
- Attach duration and illustrative outcome from DURATION_MAP / OUTCOME_MAP.
- Optional WellnessMatchingSession row if a tourist JWT is present.

C. Dosha Classifier

If tourist_profiles_training_data.xlsx is missing, a synthetic 240-row set (80 per dosha, 12% label noise, RNG seed 42) is generated so the API still boots. In the submitted tree the xlsx exists, so synthetic data is not used. Production fit uses all 102 filtered rows (no hold-out at runtime). Hyperparameters in source: n_estimators=100, random_state=42. Epochs do not apply to Random Forest. The recorded comparison CSV is treated as the offline experiment (development Step 4). A stratified 80/20 split on 102 rows implies 81 train / 21 test and reproduces five of six CSV rows in independent checks; the ANN CSV value 0.667 may not match a default MLPClassifier (max_iter=500 reproduced ~0.476). This paper therefore cites the CSV as the archived experiment and notes that the generating notebook is not in the repository.

D. Weighted Ranker

$$S = 0.35 s_{cond} + 0.20 s_{dosha} + 0.25 (q_{nlp} / 5) + 0.20 s_{budget}$$

(1)

Condition: 1.0 on substring match in conditions_text, else 0.3. Dosha: 1.0 exact, 0.6 General/Unspecified, 0.4 mismatch. Budget: 1.0 on tier substring, else 0.5. These rules are code, not a second learned model.

E. Model Selection Rationale

Random Forest was selected because it recorded the highest accuracy (0.81) and F1 (0.794) in Table II. Decision Tree was second (F1 0.705) but more brittle. SVM and k-NN collapsed (F1 0.360 and 0.332), consistent with mixed categorical encodings and a small test fold. The ranker weights were retained because ablation (Table III) shows FULL best on average quality (4.484), budget alignment (0.767), and dosha compatibility (0.513).

IV. RESULTS

TABLE II

Dosha classifier comparison (verbatim: backend/data/wellness/model_comparison_results.csv).

Model	Accuracy	Precision	Recall	F1-score
Random Forest (selected)	0.810	0.832	0.810	0.794
Decision Tree	0.714	0.702	0.714	0.705
Neural network (ANN)	0.667	0.667	0.667	0.667
Naive Bayes	0.571	0.589	0.571	0.567
SVM	0.524	0.274	0.524	0.360
k-NN	0.381	0.325	0.381	0.332

TABLE III

Matching-weight ablation (verbatim: ablation_results.json). Dosha_n is the evaluation subset size.

Configuration	Avg. quality	Budget align.	Dosha compat.	dosha_n
FULL	4.484	0.767 (76.7%)	0.513 (51.3%)	39
NO-DOSHA	4.411	0.733	0.237	38
NO-NLP-QUALITY	4.044	0.733	0.450	40
NO-BUDGET	4.443	0.433	0.476	42
CONDITION-ONLY	3.845	0.417	0.290	31

Live analytics from get_analytics(): mean nlp_quality 4.24 / 5; 4 flagged reviews on 4 centres. Duration examples: Arthritis 18-25 days (expert 21-day case); Stress/Anxiety 25-35 days; Migraine 14-30 days tourist-adapted versus a literature 90-day protocol (n=406 cited in comments). Illustrative outcome map examples: Arthritis 78%, Sciatica 80%, Skin 60%. Runtime was measured on the development workstation after a single warmup call: predict_dosha() mean 7.935 ms over 200 trials (median 7.756 ms, 95th percentile 9.055 ms, min 6.587 ms). joblib serialisation of the fitted RandomForestClassifier is 416 737 bytes (406.97 KB). Ranking 69 Ayurveda centres with match_centers() mean 1.276 ms over 100 trials (median 1.217 ms, 95th percentile 1.820 ms). These are CPU times in CPython, not GPU. A single-quiz predict_proba peak of 94.0% on one Vata example is a confidence score, not hold-out accuracy; classifier accuracy remains 0.81 from Table II.

TABLE IV

Measured runtime and artefact size (development PC, scikit-learn 1.5.1, n_estimators=100).

Quantity	Value	Protocol
RF features / trees / train rows	6 / 100 / 102	matching_engine.py import fit
predict_dosha() mean	7.94 ms	200 trials after 1 warmup
predict_dosha() median / p95	7.76 ms / 9.06 ms	same run
RF joblib size	406.97 KB (416 737 B)	joblib.dump of DOSHA_MODEL
match_centers() mean (Ayurveda, top_n=6)	1.28 ms	100 trials, 69-centre pool

TABLE V

Selected duration / outcome lookups (matching_engine.py). Outcome % is not a clinical claim.

Condition	Duration range	Illustrative %	Source note in code
Arthritis	18-25 days	78	Domain expert 21-day case; n=406 migraine study as reference class
Sciatica	18-25 days	80	1 documented full-recovery case + literature pattern
Hypertension	12-18 days	68	Ayurveda Sarana Beach Hospital market case
Detox / weight	10-16 days	72	Heritage Ayurveda 14 days / 5 kg report
Migraine	14-30 days (adapted)	71	Literature protocol n=406, 90-day course
General wellness	3-7 days	85	Subjective relaxation, high self-report

V. DISCUSSION

A. Contribution to Tour Ceylon

Wellness is the only module with a trained classifier and a recorded model bake-off. It turns the platform from plan-a-route-and-call-for-help into personalised health-and-retreat discovery, using the same tourist JWT when present. Admin analytics surface Table II-III so a panel can inspect the same numbers as this report.

B. Limitations

- Only 6 real Vikruti cases; 104 rows are expert-rule augmentations (70/30 dominant/overlap).
- Production RF is fit on all 102 rows; CSV metrics are an offline hold-out artifact.
- 50 of 83 centres have unspecified dosha focus, so `s_dosha` is often 0.6 by default.
- Risk detection is a keyword list (unhygienic, scam, unsafe, ...), not a text classifier.
- OOV quiz answers silently become the first LabelEncoder class.
- Spiritual path still predicts dosha, which may be weakly relevant.
- Outcome percentages must not be quoted as treatment efficacy in a medical sense.
- Training script/notebook not versioned; ANN CSV row may be irreproducible with default MLP.
- Admin authentication is a shared password, not per-user RBAC.

C. Future Work

Nested cross-validation on a larger real labelled set; class-weighting for Kapha (16 rows); separate spiritual ranker without dosha; transformer review NLP; logged inference latency; serialize the fitted RF to disk instead of refitting at every process start.

VI. CONCLUSION

The wellness component is a hybrid system: Random Forest dosha classification (`n_estimators=100`, recorded F1 0.794) and a four-term linear centre ranker (full ablation quality 4.484 vs 3.845 condition-only) over 83 centres. These are the numbers that belong in an IEEE results section. Itinerary and SOS, by contrast, are heuristic and geospatial respectively and must not inherit these F1 scores.

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- [5] Domain expert case set labelled Ayya's Case (6 real profiles) and expert-rule augmentation (104 rows, 70/30 dominant/overlap) in `tourist_profiles_training_data.xlsx`.
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- [11] Tour Ceylon, `backend/data/wellness/model_comparison_results.csv` and `ablation_results.json`, 2026.
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Disclaimer: Treatment durations and outcome percentages are decision-support illustrations for a tourism information system. They are not medical advice or validated clinical endpoints.